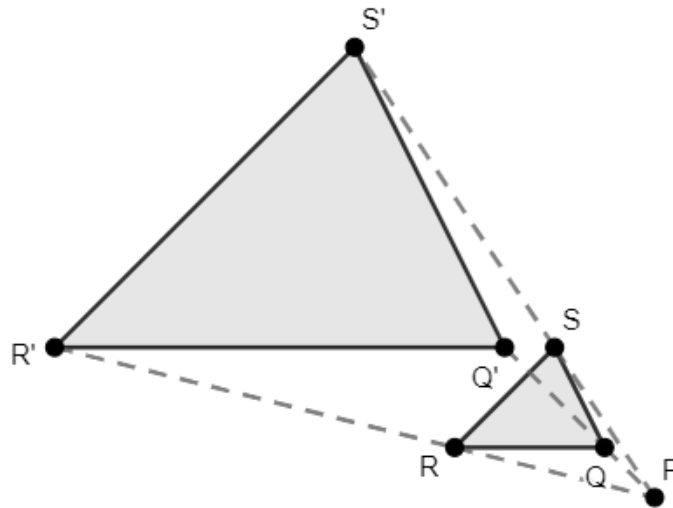


$\triangle Q'R'S'$ was created by dilating $\triangle QRS$. The length of $\overline{QR}$ is 3 and the length of $\overline{Q'R'}$ is 9.



1. $\triangle QRS$ is the ☐ image ☒ preimage, $\triangle Q'R'S'$ is the ☒ image ☐ preimage, and the center of

dilation is point ☒ P ☐ Q ☐ Q'.

2. Determine the dilation ratio.

$$\frac{Q'R'}{QR} = \frac{9}{3} = 3$$

3. If the length of $\overline{Q'S'}$ is $3\sqrt{5}$, determine the length of $\overline{QS}$.
 $QS = \sqrt{5}$

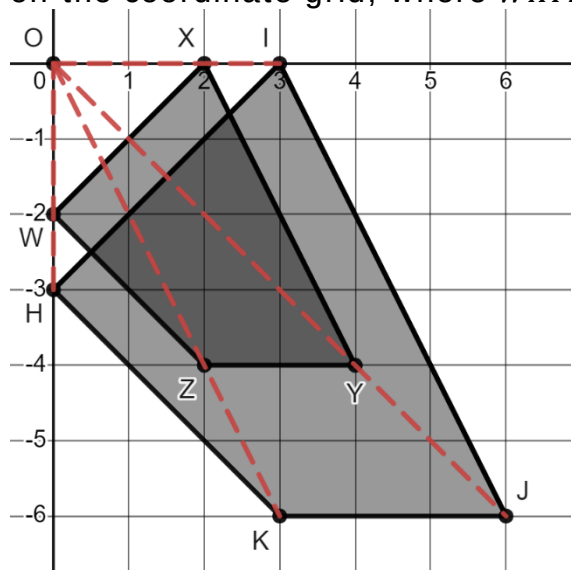
4. What is the ratio of $\frac{RS}{R'S'}$?

$$\frac{RS}{R'S'} = \frac{1}{3}$$

5. What is the ratio of $\frac{P'R'}{PR}$?

$$\frac{P'R'}{PR} = \frac{3}{1} = 3$$

$WXYZ$ and $HIJK$ are shown on the coordinate grid, where $WXYZ$ is a dilation of $HIJK$.



6. Complete the table.

Segment	Slope	Distance
$\overline{YZ}$	0	2
$\overline{JK}$	0	3
$\overline{WX}$	$\frac{2}{2} = 1$	$\sqrt{8} = 2\sqrt{2}$
$\overline{HI}$	$\frac{3}{3} = 1$	$\sqrt{18} = 3\sqrt{2}$

7. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.

8. Write a description of the dilation by giving the center of dilation and the scale factor.

$(x,y) \rightarrow (\frac{2}{3}x, \frac{2}{3}y)$
Center of dilation is (0,0) and scale factor is $\frac{2}{3}$

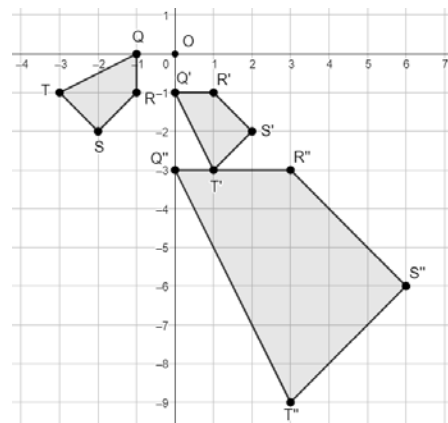
9. What is the ratio $\frac{OX}{OI}$?

$\frac{OX}{OI} = \frac{2}{3}$

10. What is the ratio $\frac{OX}{XI}$?

$\frac{OX}{XI} = \frac{2}{1}$

A sequence of transformations of $QRST$ is shown on the coordinate grid. $QRST$ is rotated then dilated, where $Q'R'S'T'$ is the rotation and $Q''R''S''T''$ is the dilation of $Q'R'S'T'$.



Complete the descriptions:

- $QRST$ was rotated 90° counterclockwise to create $Q'R'S'T'$.
- $Q'R'S'T'$ was dilated using a scale factor of 3 centered at (0,0) to create $Q''R''S''T''$.

3. Complete the table:

Transformation	Algebraic Description
Rotation	$(x, y) \rightarrow (-y, x)$
Dilation	$(x, y) \rightarrow (3x, 3y)$

4. Determine the coordinates if $QRST$ is first dilated then rotated:

Preimage $QRST$	
Q	$(-1, 0)$
R	$(-1, -1)$
S	$(-2, -2)$
T	$(-3, -1)$

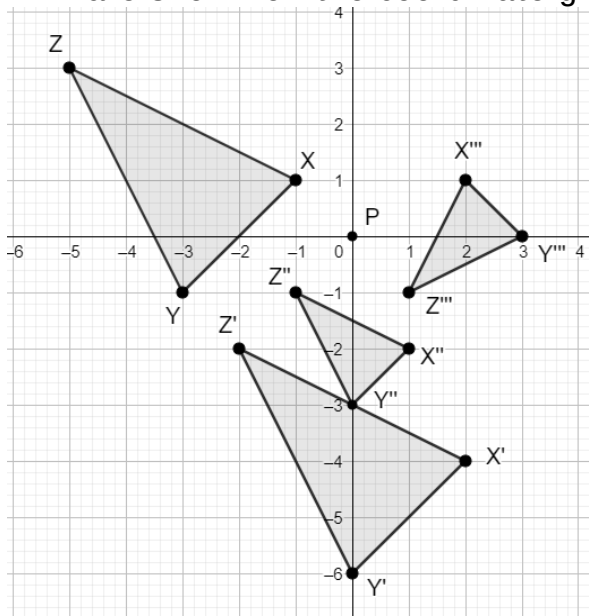
Dilation $Q'R'S'T'$	
Q'	$(-3, 0)$
R'	$(-3, -3)$
S'	$(-6, -6)$
T'	$(-9, -3)$

Rotation $Q''R''S''T''$	
Q''	$(0, -3)$
R''	$(3, -3)$
S''	$(6, -6)$
T''	$(3, -9)$

- Based on the results above, the sequence of transformations commutative.

☒ is
☐ is not

$XYZ, X'Y'Z', X''Y''Z'',$ and $X'''Y'''Z'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- XYZ is first

☒ translated
☐ reflected
☐ rotated
☐ dilated

 to $X'Y'Z'$, then

☐ translated
☐ reflected
☐ rotated
☒ dilated

 to $X''Y''Z''$,
- and finally

☐ translated
☐ reflected
☒ rotated
☐ dilated

 to $X'''Y'''Z'''$.
7. Describe the transformation for $XYZ \rightarrow X'Y'Z'$.
Translate 5 units down, 3 units right
8. Describe the transformation for $X'Y'Z' \rightarrow X''Y''Z''$.
Dilate with a scale factor of $\frac{1}{2}$
9. Describe the transformation for $X''Y''Z'' \rightarrow X'''Y'''Z'''$.
Rotate 90° counterclockwise

10. Write the algebraic description for the sequence of transformations from 7-9:

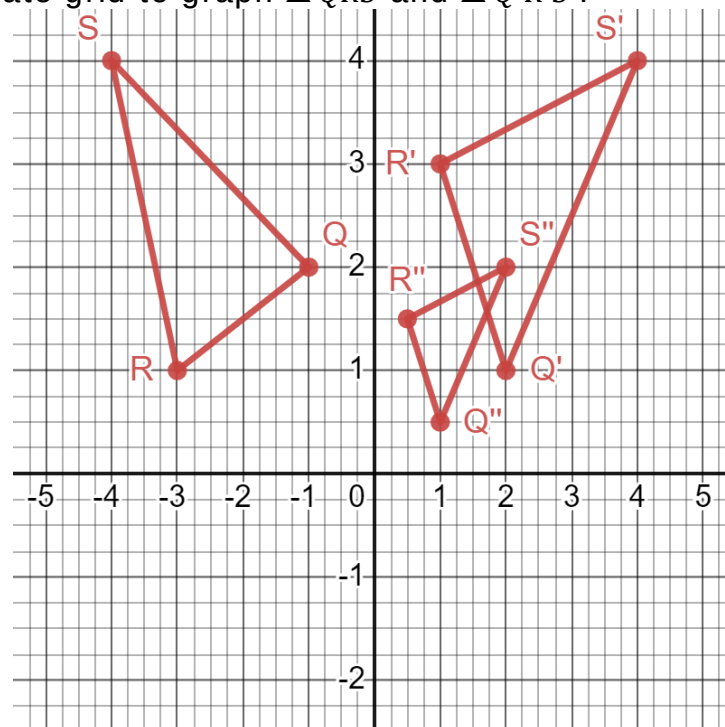
Transformation	Algebraic Description
Translation	$(x,y) \rightarrow (x+3,y-5)$
Dilation	$(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$
Rotation	$(x,y) \rightarrow (-y,x)$

The coordinates of $\triangle QRS$ and $\triangle Q'R'S'$ are shown in the table, where $\triangle QRS$ is the preimage:

$\triangle QRS$		
Vertex	x	y
Q	-1	2
R	-3	1
S	-4	4

$\triangle Q'R'S'$		
Vertex	x	y
Q'	2	1
R'	1	3
S'	4	4

1. Use the coordinate grid to graph $\triangle QRS$ and $\triangle Q'R'S'$.

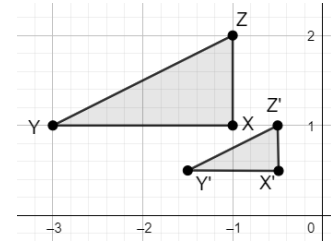


2. What type of transformation was applied to the preimage $\triangle QRS$?
Rotation 90° clockwise
3. Was distance preserved in this transformation?
Yes
4. $\triangle Q'R'S'$ is further transformed by dilating with a scale factor of $\frac{1}{2}$ centered at the origin to create $\triangle Q''R''S''$. Graph $\triangle Q''R''S''$ on the coordinate grid above.
 $Q'' = (1, 0.5)$, $R'' = (0.5, 1.5)$, $S'' = (2, 2)$
5. Was distance preserved in this second transformation from $\triangle Q'R'S'$ to $\triangle Q''R''S''$?
No

Complete the following statements about the transformation of preimage XYZ :

6. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☒ dilation



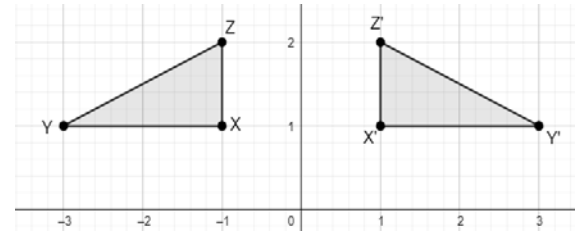
This transformation

- ☐ does
- ☒ does not

preserve distance.

7. Image $X'Y'Z'$ was created by

- ☐ translation
- ☒ reflection
- ☐ rotation
- ☐ dilation



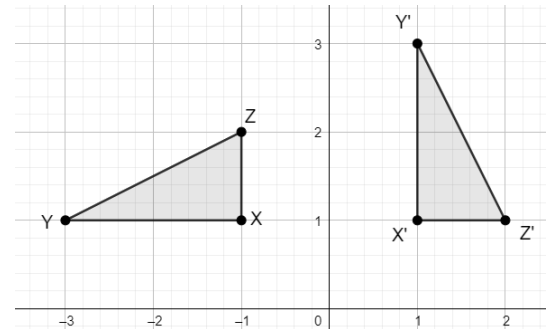
This transformation

- ☒ does
- ☐ does not

preserve distance.

8. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☒ rotation
- ☐ dilation



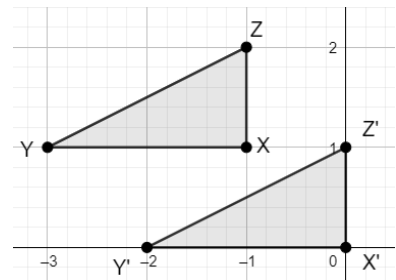
This transformation

- ☒ does
- ☐ does not

preserve distance.

9. Image $X'Y'Z'$ was created by

- ☒ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation



This transformation

- ☒ does
- ☐ does not

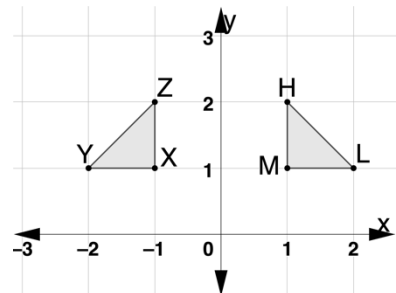
preserve distance.

10. Image MLH was created either by

or by

- ☐ translation
- ☐ reflection
- ☒ rotation
- ☐ dilation

- ☐ translation
- ☒ reflection
- ☐ rotation
- ☐ dilation

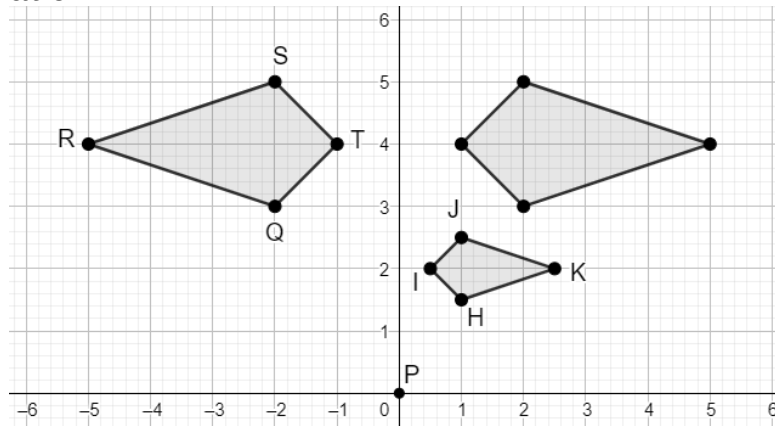


Both of these transformations

- ☒ do
- ☐ do not

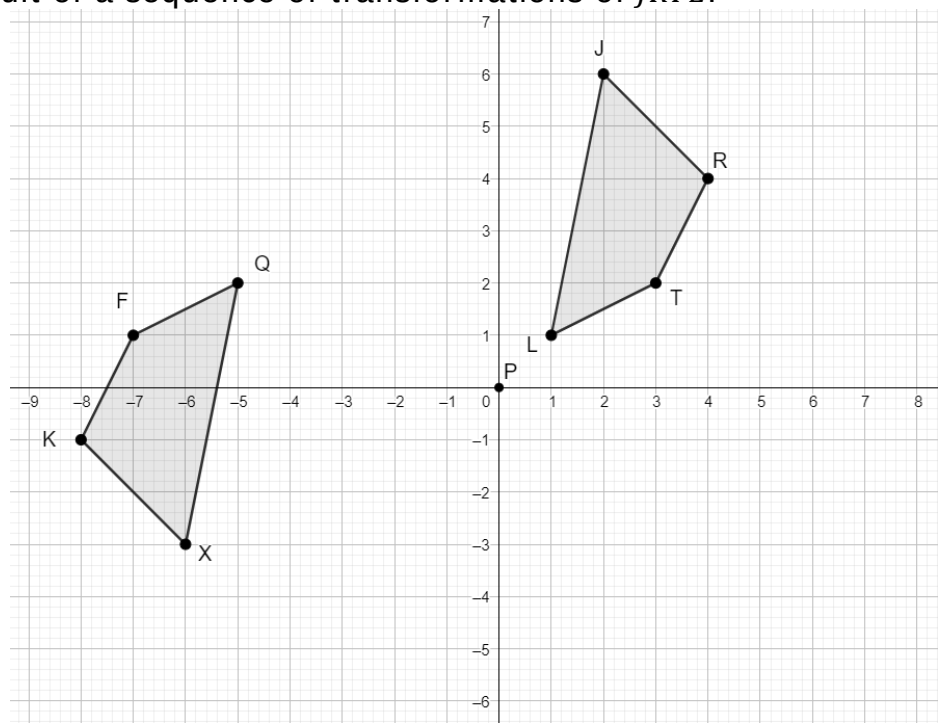
preserve distance.

$HIJK$ is the result of a sequence of transformations of $QTSR$. The unlabelled quadrilateral is the first transformation.



- Describe the first transformation of $QTSR$.
Reflection over y -axis
- Write the transformation algebraically.
 $(x, y) \rightarrow (-x, y)$
- Describe the second transformation that results in $HIJK$.
Dilation with a scale factor of $\frac{1}{2}$ centered at $(0, 0)$
- Write the second transformation algebraically.
 $(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$
- Explain if the sequence of transformations preserves distance.
The first transformation does preserve distance, but the second transformation does not.

$XKFQ$ is the result of a sequence of transformations of $JRTL$.



Julia and Nathan were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

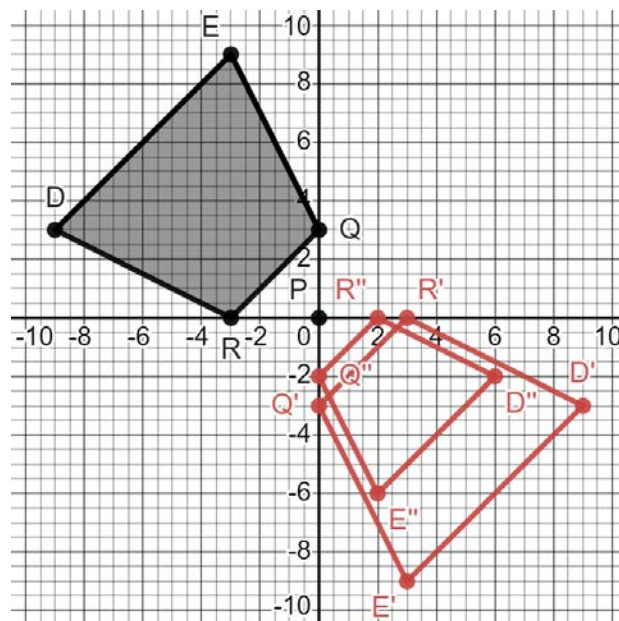
			First Transformation	Second Transformation
6.	Julia's Sequence	Written Description	Translate right 4 and down 3	Rotate 180° about the origin.
7.		Algebraic Description	$(x, y) \rightarrow (x + 4, y - 3)$	$(x, y) \rightarrow (-x, -y)$
8.	Nathan's Sequence	Written Description	Rotate 180° about the origin.	Translate right 4 and down 3
9.		Algebraic Description	$(x, y) \rightarrow (-x, -y)$	$(x, y) \rightarrow (x + 4, y - 3)$

10. Considering both proposed sequences, you conclude that correct.

- ✖ Julia's sequence is
- ☐ Nathan's sequence is
- ☐ Both sequences are

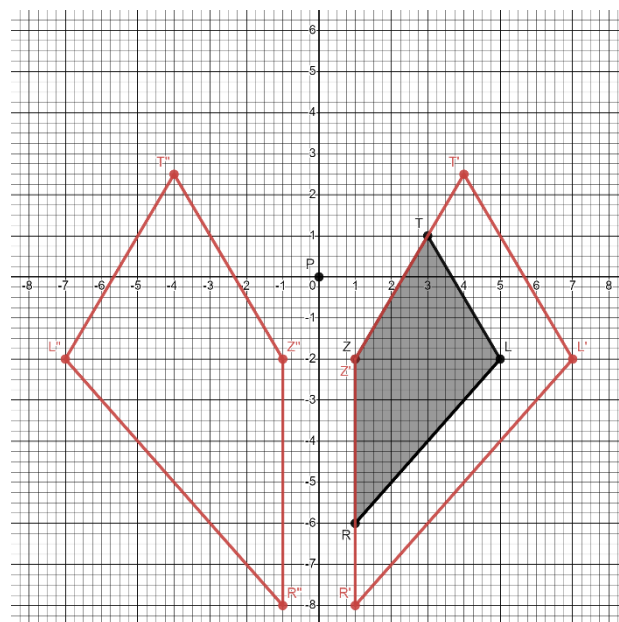
$QEDR$ and point P are shown on the coordinate plane.

1. Draw the quadrilateral that is a 180° rotation of $QEDR$ about the origin. Label the quadrilateral $Q'E'D'R'$.
2. Draw the quadrilateral that is a dilation of $Q'E'D'R'$ with a center of dilation at point P and a scale factor of $\frac{2}{3}$. Label the quadrilateral $Q''E''D''R''$.



$ZTLR$ and point P are shown on the coordinate plane.

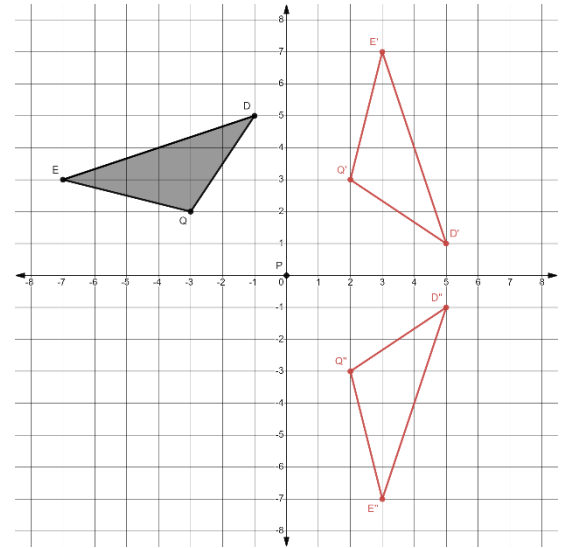
3. Draw the quadrilateral that is a dilation of $ZTLR$ with a center of dilation at point Z and a scale factor of $\frac{3}{2}$. Label the quadrilateral $Z'T'L'R'$.
4. Draw the quadrilateral that is a reflection of $Z'T'L'R'$ across the y -axis. Label the quadrilateral $Z''T''L''R''$.



$\triangle QED$ and point P are shown on the coordinate plane.

5. Draw the triangle that is a 90° clockwise rotation of $\triangle QED$ about point P . Label the quadrilateral $\triangle Q'E'D'$.

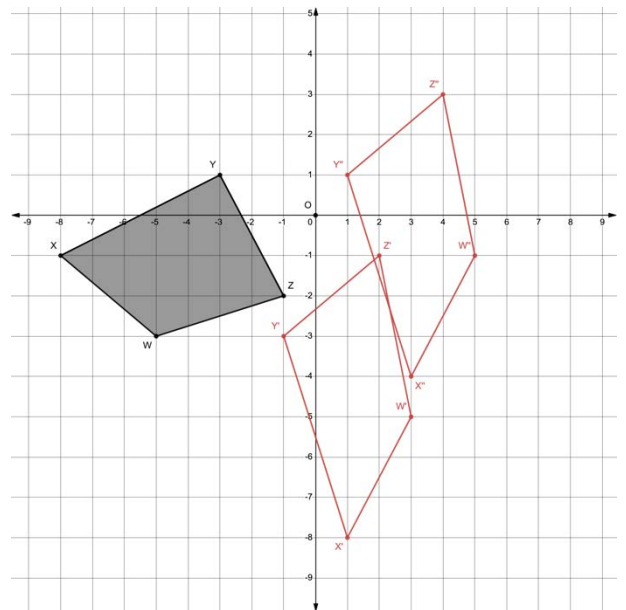
6. Draw the triangle that is a reflection of $\triangle Q'E'D'$ across the x -axis. Label the quadrilateral $\triangle Q''E''D''$.



$WXYZ$ and point O are shown on the coordinate plane.

7. Draw the quadrilateral that is a 90° counterclockwise rotation of $WXYZ$ about the origin. Label the quadrilateral $W'X'Y'Z'$.

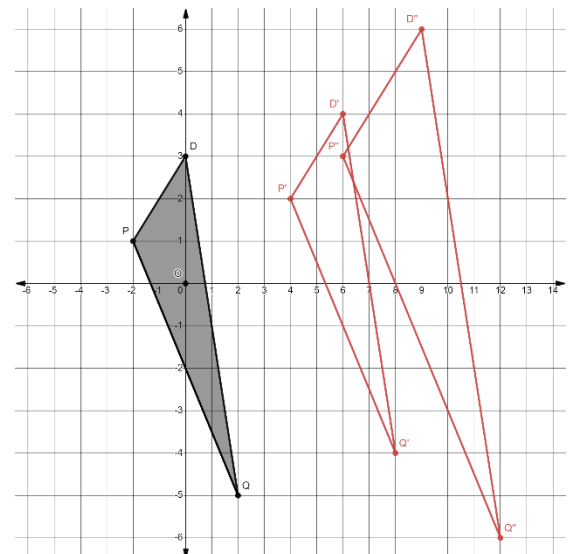
8. Draw the quadrilateral that is a translation of $W'X'Y'Z'$ 2 units right and 4 units up. Label the quadrilateral $W''X''Y''Z''$.



$\triangle PDQ$ and point O are shown on the coordinate plane.

9. Draw $\triangle P'D'Q'$ as a result of $(x, y) \rightarrow (x + 6, y + 1)$.

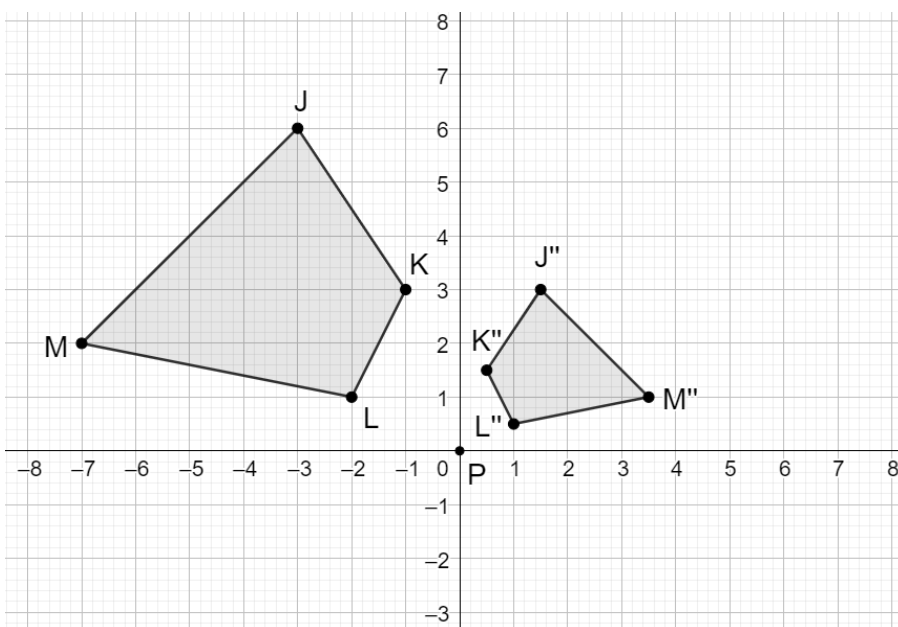
10. Draw $\triangle P''D''Q''$ as a result of $(x, y) \rightarrow (\frac{3}{2}x, \frac{3}{2}y)$.



Geometry
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1. When a sequence of transformations results in both angles and sides of the image that are congruent to corresponding angles and sides of the preimage, the sequence ☐ does ☒ does not include a dilation. If it ☒ does ☐ does not include a dilation, the corresponding ☒ angles ☐ sides are congruent while the corresponding ☐ angles ☒ sides are proportional.



2. Write a sequence of transformations for preimage $JKLM$ that would show whether the figures are similar.

Reflection over y -axis and dilation with scale factor $\frac{1}{2}$

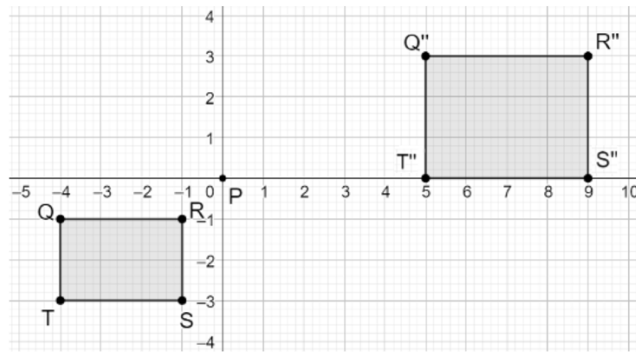
$$(x, y) \rightarrow (-x, y) \text{ and } (x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$$

3. Is $J''K''L''M''$ similar to the preimage?

☒ Yes
☐ No

4. Explain the reason for your answer for #3.

Angles are congruent and sides are proportional.



5. Write a sequence of transformations for preimage $QRST$ that would show whether the figures are similar.

Figures are not similar.

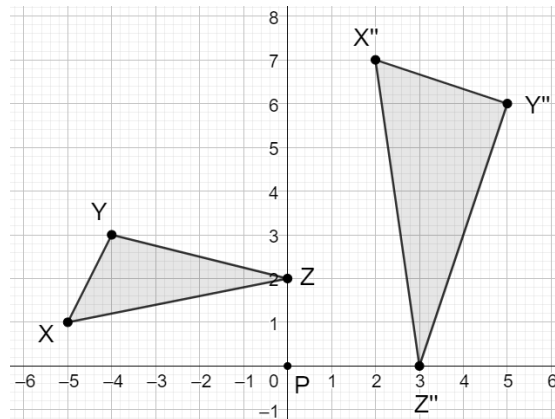
6. Is $Q''R''S''T''$ similar to the preimage?

☐ Yes

☒ No

7. Explain the reason for your answer for #6.

The sides are not proportional.



8. Write a sequence of transformations for preimage $\triangle XYZ$ that would show whether the figures are similar.

Figures are not similar.

9. Is $\triangle X''Y''Z''$ similar to the preimage?

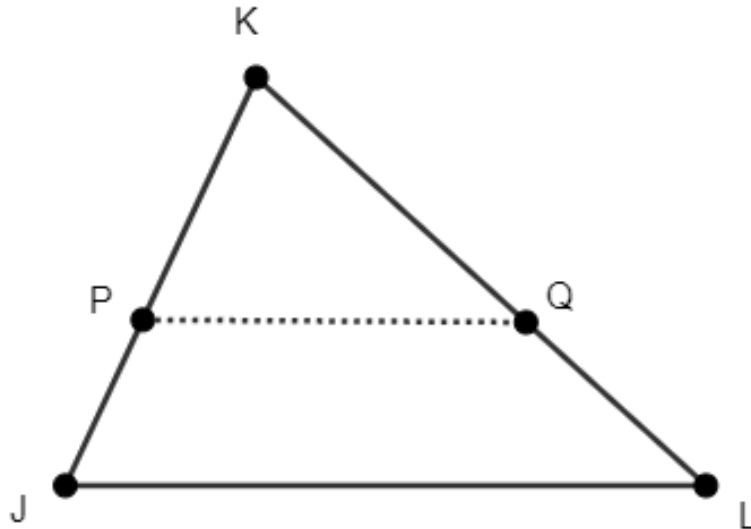
☐ Yes

☒ No

10. Explain the reason for your answer for #9.

The sides are not proportional.

$\triangle KJL$ is shown, where $m\angle KJL = 65^\circ$, $m\angle KLJ = 42^\circ$, and $m\angle JKL = 73^\circ$. Point P is on $\overline{KJ}$ and Point Q is on $\overline{KL}$. $\overline{PQ}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle JKL$		Measurements of $\triangle PKQ$	
$JK = 6 \text{ cm}$	$m\angle KJL = 65^\circ$	$PK = 3.5 \text{ cm}$	$m\angle KPQ = 65^\circ$
$KL = 8 \text{ cm}$	$m\angle JKL = 73^\circ$	$KQ = 4.6 \text{ cm}$	$m\angle PKQ = 73^\circ$
$JL = 8.5 \text{ cm}$	$m\angle KLJ = 42^\circ$	$PQ = 5 \text{ cm}$	$m\angle KQP = 42^\circ$

2. The corresponding angles of $\triangle KJL$ and $\triangle KPQ$ are $\angle K$ and $\angle K$, $\angle J$ and $\angle P$, $\angle L$ and $\angle Q$.

3. Complete the table.

Sides	Ratio
Determine the ratio of JK to PK .	$\frac{6}{3.5} = 1.714$
Determine the ratio of KL to KQ .	$\frac{8}{4.6} = 1.739$
Determine the ratio of JL to PQ .	$\frac{8.5}{5} = 1.700$

4. The corresponding sides of $\triangle KJL$ and $\triangle KPQ$ are $\overline{JK}$ and $\overline{PK}$, $\overline{KL}$ and $\overline{KQ}$, $\overline{JL}$ and $\overline{PQ}$.

5. $\triangle KJL$ is similar to $\triangle KPQ$.
6. Describe a sequence of transformations that can be used to map $\triangle KJL$ onto $\triangle KPQ$.
 $\frac{1}{1.7} \approx 0.59$
 Dilation with K as the center of dilation and a scale factor of 0.59.

Complete each relationship for $\triangle KJL$ and $\triangle KPQ$.

7. $\frac{PK}{JK} = \frac{PQ}{JL}$

8. $\frac{JK}{PK} = \frac{LJ}{QP}$

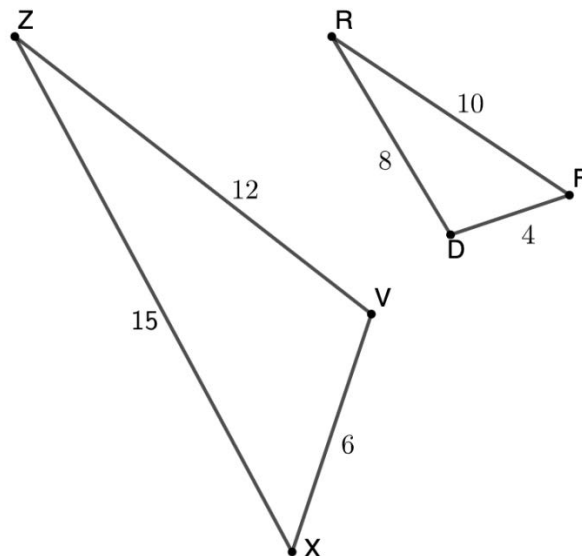
9. $\frac{KQ}{KL} = \frac{KP}{KJ}$

10. Observing that $\frac{JL}{PQ} \neq \frac{LQ}{QK}$, your classmate Moriah claims that $\triangle KJL$ and $\triangle KPQ$ are not similar. If her logic is faulty, explain why.
 $\overline{LQ}$ is a portion of one side of $\triangle KJL$ and not a corresponding side to $\overline{QK}$, so her logic is faulty.

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$\triangle ZVX$ and $\triangle RFD$ are shown.



1. Complete the table.

Measurements of $\triangle ZVX$	Measurements of $\triangle RFD$	Ratio
$ZX = 15$	$RF = 10$	$\frac{ZX}{RF} = \frac{15}{10} = \frac{3}{2}$
$XV = 6$	$FD = 4$	$\frac{XV}{FD} = \frac{6}{4} = \frac{3}{2}$
$VZ = 12$	$DR = 8$	$\frac{VZ}{DR} = \frac{12}{8} = \frac{3}{2}$

2. What similarity theorem can be applied to show that the triangles are similar?
Side-Side-Side Similarity

3. Write the similarity statement.
 $\triangle ZVX \sim \triangle RFD$

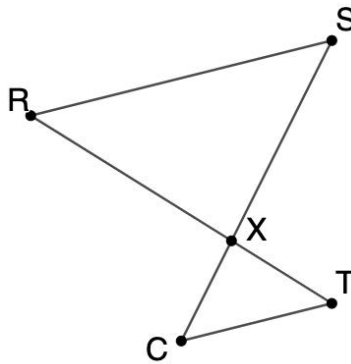
4. Complete each.

$$\frac{ZV}{XV} = \frac{RD}{FD}$$

$$\frac{RF}{ZX} = \frac{DF}{VX}$$

$$\frac{XZ}{VZ} = \frac{FR}{DR}$$

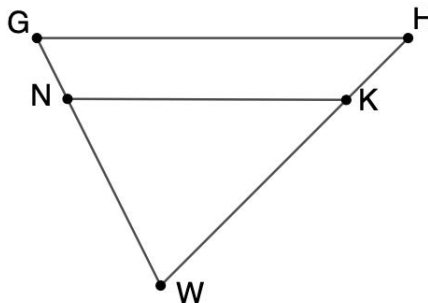
$\triangle RSX$ and $\triangle XTC$ are shown where $\angle RSX \cong \angle TCX$.



Complete the two-column proof.

	Statement	Reason
	$\angle RSX \cong \angle TCX$	Given
5.	$\angle SXR \cong \angle CXT$	Vertical angles are congruent
6.	$\triangle RSX \sim \triangle \underline{\hspace{2cm}}$	AA ~

$\triangle GWH$ and $\triangle NWK$ are shown.



Given: $WN = \frac{2}{3}WG$ and $WK = \frac{2}{3}WH$

Prove: $\triangle GWH \sim \triangle NWK$

	Statement	Reason
	$WN = \frac{2}{3}WG$ and $WK = \frac{2}{3}WH$	Given
7.	$\underline{\hspace{1cm}} = \frac{2}{3}$ and $\underline{\hspace{1cm}} = \frac{2}{3}$	Division Property of Equality
8.	$\frac{WN}{WG} = \frac{WK}{WH}$	Transitive Property of Equality
9.	$\angle GWH \cong \angle NWK$	Reflexive Property of Congruence
10.	$\triangle GWH \sim \triangle NWK$	SAS ~

Geometry
Unit 6
Lesson 9 Homework

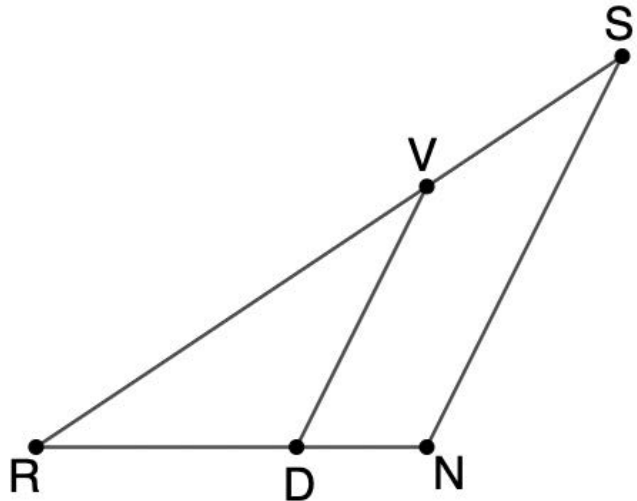
Name Answer Key
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$\triangle RSN$ with $\overline{VD}$ is shown.

Given: $RV = \frac{2}{3}RS$ and
 $RD = \frac{2}{3}RN$

Prove: $\triangle RSN \sim \triangle RVD$

Complete the two-column proof with the provided statements and reasons.



Statement	Reason
$RV = \frac{2}{3}RS$ and $RD = \frac{2}{3}RN$	Given
1. $\frac{RV}{RS} = \frac{2}{3}$ and $\frac{RD}{RN} = \frac{2}{3}$	Division Property of Equality
2. $\frac{RV}{RS} = \frac{RD}{RN}$	Transitive Property of Equality
3. $\angle VRD \cong \angle SRN$	Reflexive Property of Congruence
4. $\triangle RSN \sim \triangle RVD$	SAS ~

$\triangle RSN \sim \triangle RVD$

SAS ~

$\frac{RV}{RS} = \frac{RD}{RN}$

Transitive Property of Equality

$\angle VRD \cong \angle SRN$

Division Property of Equality

$\frac{RV}{RS} = \frac{2}{3}$ and $\frac{RD}{RN} = \frac{2}{3}$

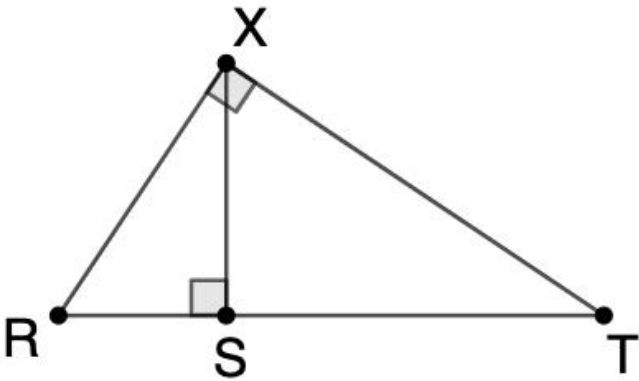
Reflexive Property of Congruence

$\triangle RXT$ with $\overline{XS}$ is shown.

Given: $\overline{XS} \perp \overline{RT}$ and
 $\overline{RX} \perp \overline{XT}$

Prove: $\triangle XTR \sim \triangle SXR$

Complete the two-column proof with the provided statements and reasons.

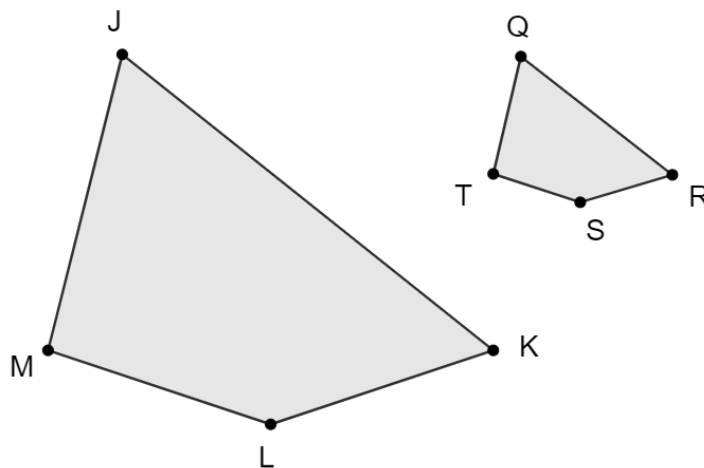


	Statement	Reason
	$\overline{XS} \perp \overline{RT}$ and $\overline{RX} \perp \overline{XT}$	Given
5.	$\angle XSR$ and $\angle TXR$ are right angles	Definition of perpendicular lines
6.	$m\angle XSR = 90^\circ$ and $m\angle TXR = 90^\circ$	Definition of right angles
7.	$m\angle XSR = m\angle TXR$	Transitive Property of Equality
8.	$\angle XSR \cong \angle TXR$	Definition of congruence
9.	$\angle XRS \cong \angle TRX$	Reflexive Property of Congruence
10.	$\triangle XTR \sim \triangle SXR$	AA ~

$\angle XSR \cong \angle TXR$	AA ~
$\triangle XTR \sim \triangle SXR$	Definition of congruence
$\angle XRS \cong \angle TRX$	Definition of perpendicular lines
$m\angle XSR = m\angle TXR$	Reflexive Property of Congruence
$m\angle XSR = 90^\circ$ and $m\angle TXR = 90^\circ$	Transitive Property of Equality
$\angle XSR$ and $\angle TXR$ are right angles	Definition of right angles

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Fill in the table given that $JKL \sim QRS$.

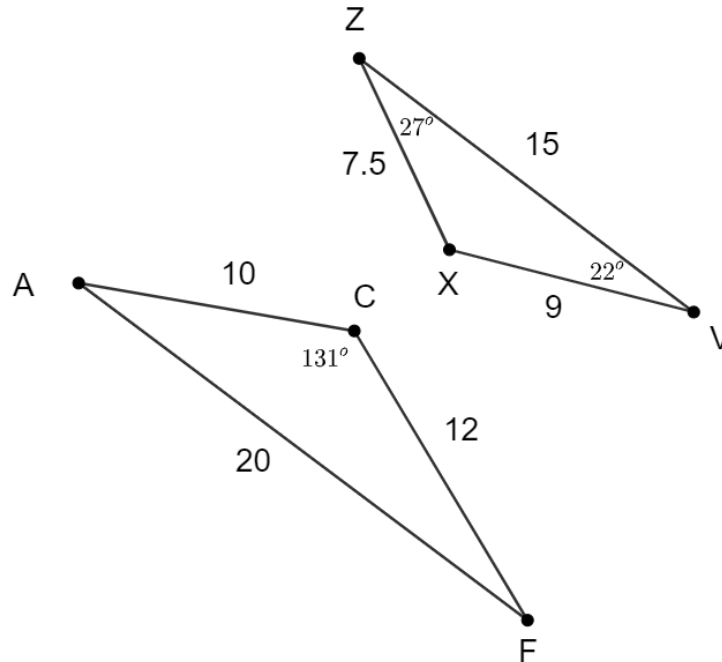
	Corresponding Angles	Corresponding Sides
1.	$\angle JML \cong \angle QTS$	$\frac{JM}{QT} = \frac{ML}{TS}$
2.	$\angle TQR \cong \angle MJK$	$\frac{LK}{SR} = \frac{JK}{QR}$
3.	$\angle TSR \cong \angle MLK$	$\frac{JK}{QR} = \frac{JM}{QT}$
4.	$\angle JKL \cong \angle QRS$	$\frac{ML}{TS} = \frac{LK}{SR}$

5. If $ML = 6.5$, $TS = 2.5$, and $QR = 5$, determine JK .

$$\frac{6.5}{2.5} = \frac{JK}{5}$$

$$JK = 13$$

$\triangle ACF$ and $\triangle VXZ$ are shown. $AC = 10$, $CF = 12$, $AF = 20$, $VX = 9$, $XZ = 7.5$, $ZV = 15$, $m\angle ACF = 131^\circ$, $m\angle XZV = 27^\circ$, and $m\angle ZVX = 22^\circ$.



6. What similarity theorem or postulate could be used to prove the triangles are similar?

Answers may vary. SAS ~ Theorem using $\frac{AC}{ZX} = \frac{CF}{XV}$ and $\angle C \cong \angle X$ or SSS.

7. Write a similarity statement for $\triangle ACF$ and $\triangle VXZ$.

$\triangle FCA \sim \triangle VXZ$

Complete the table.

	Angle Measures	Side Proportions
8.	$m\angle CAF = 27^\circ$	$\frac{AC}{ZX} = \frac{4}{3}$
9.	$m\angle CFA = 22^\circ$	$\frac{CF}{XV} = \frac{4}{3}$
10.	$m\angle ZXV = 131^\circ$	$\frac{FA}{VZ} = \frac{4}{3}$

Geometry

Unit 6

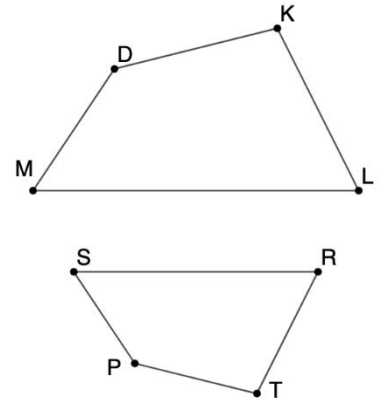
Lesson 11 Homework

Name Answer Key

Date _____

Period _____

$DKLM$ and $PTRS$ are shown, where $DKLM \sim PTRS$ with $MD = 20$, $DK = 6z + 6$, $KL = 8w + 1$, $LM = 40$, $PT = 12$, $TR = 4w - 2$, and $RS = 16$.



- Determine the value of each variable.

$$w = 3$$

$$z = 4$$

- Determine the value of each side.

$$DK = 30$$

$$SP = 8$$

$$TR = 10$$

- What is the perimeter of $DKLM$?

$$\text{Perimeter} = 30 + 25 + 40 + 20 = 115$$

Consider additional measurements of angles for $DKLM$ and $PTRS$ where $m\angle DKL = 150^\circ$, $m\angle MDK = (15x - 25)^\circ$, $m\angle SPT = (12x + 5)^\circ$, $m\angle DML = (5y - 5)^\circ$, $m\angle PSR = (4y + 4)^\circ$.

- Determine the value of each variable.

$$x = 10$$

$$y = 9$$

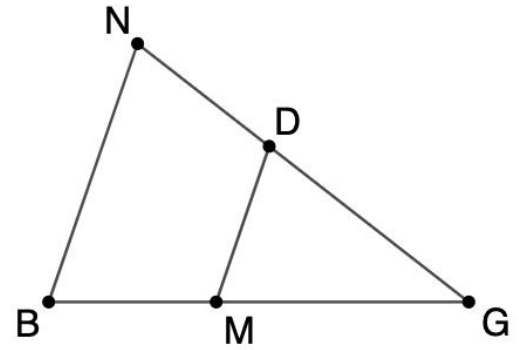
- Determine the value of each angle.

$$m\angle MDK = 125^\circ$$

$$m\angle PTR = 150^\circ$$

$$m\angle PSR = 40^\circ$$

$\triangle NBG$ and $\triangle DMG$ are shown, where $\triangle NBG \sim \triangle DMG$ with $DM = 8$, $NB = 32$, $DG = 7$, $MG = 2y - 8$, and $BG = 4y + 4$.



6. What is the scale factor from $\triangle DMG$ to $\triangle NBG$?

$$\frac{32}{8} = 4$$

7. What is the value of the variable, y ?

$$\begin{aligned} \frac{32}{8} &= \frac{4y+4}{2y-8} \\ 32(2y-8) &= 8(4y+4) \\ y &= 9 \end{aligned}$$

8. What is the length of BG ?

$$4(9) + 4 = 40 = BG$$

Consider additional measurements of angles for $\triangle NBG$ and $\triangle DMG$ where $m\angle GNB = (8z + 3)^\circ$, $m\angle DMG = (7w - 9)^\circ$, $m\angle NBG = (6w + 3)^\circ$, $m\angle GDM = (10z - 15)^\circ$.

9. Determine the value of each variable.

$$w = 12$$

$$z = 9$$

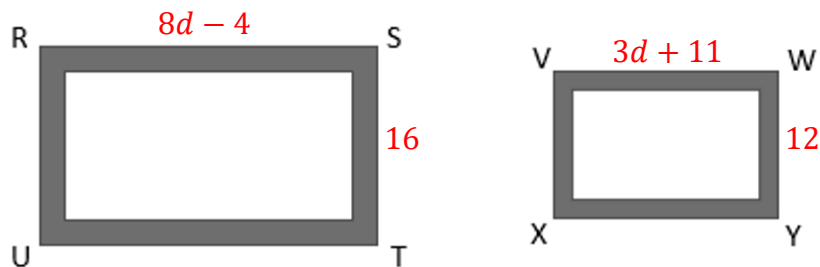
10. Determine the value of each angle.

$$m\angle GNB = 75^\circ$$

$$m\angle GBN = 75^\circ$$

$$m\angle DGM = 30^\circ$$

Hillary is custom building frames for a set of family photos. She has designed two similar rectangular frames, $RSTU$ and $VWYX$. The diagram shows the two different sized frames, where $WY = 12$ inches (in.), $VW = (3d + 11)$ in., $ST = 16$ in., and $RS = (8d - 4)$ in.



1. What is the value of d ?

$$\frac{8d-4}{3d+11} = \frac{16}{12}$$

$$12(8d - 4) = 16(3d + 11)$$

$$96d - 48 = 48d + 176$$

$$48d = 224$$

$$d = 4\frac{2}{3}$$

2. What is the area of $RSTU$? Include units.

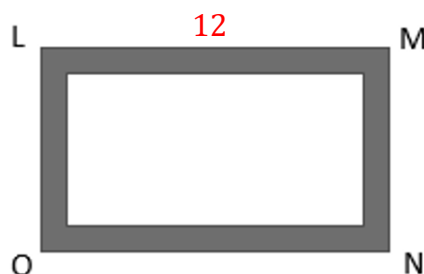
$$RS = 8\left(4\frac{2}{3}\right) - 4 = 33\frac{1}{3} \quad \text{Area} = \left(33\frac{1}{3}\right)(16) = 533\frac{1}{3} \text{ in}^2$$

3. What is the perimeter of $VWYX$? Include units.

$$VW = 3\left(4\frac{2}{3}\right) + 11 = 25$$

$$P = 2(25) + 2(12) = 74 \text{ in}$$

Hillary created a third similar rectangular frame, $LMNO$, shown below, where $LM = 12$ in.



4. What is the length of $\overline{LO}$? Include units.

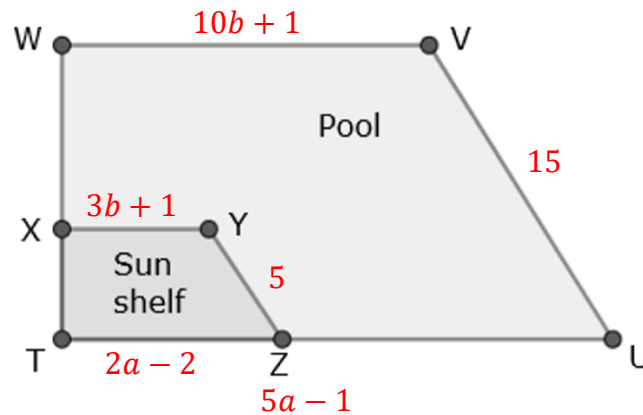
$$\frac{25}{12} = \frac{12}{LO}$$

$$LO = \frac{(12)(12)}{25} = 5.76 \text{ in}$$

5. What is the perimeter, in inches, of $LMNO$?

$$P = 2(12) + 2(5.76) = 35.52 \text{ in}$$

Laurel and her family are having a pool built in the backyard. The pool, $TUVW$, is shaped like a trapezoid, with a smaller similar trapezoid, $TZYX$, for a sun shelf. A diagram of the pool is shown, where $TU = (5a - 1)$ feet (ft), $TZ = (2a - 2)$ ft, $ZY = 5$ ft, $UV = 15$ ft, $VW = (10b + 1)$ ft, and $YX = (3b + 1)$ ft.



6. Determine the value of each variable:

$$\begin{array}{lcl} a = \underline{5} & \frac{15}{5} = \frac{5a-1}{2a-2} & \frac{15}{5} = \frac{10b+1}{3b+1} \\ b = \underline{2} & 15(2a-2) = 5(5a-1) & 15(3b+1) = 5(10b+1) \\ & 30a-30 = 25a-5 & 45b+15 = 50b+5 \\ & 5a = 25 & 5b = 10 \\ & a = 5 & b = 2 \end{array}$$

7. Determine the value of each side:

$$\begin{array}{lcl} WV = \underline{21} & WV = 10(2) + 1 = 21 \\ TZ = \underline{8} & TZ = 2(5) - 2 = 8 \end{array}$$

8. If $WV = WT$, what is the perimeter, in feet, of the pool?

$$P = 21 + 21 + 15 + [5(5) - 1] = 81 \text{ ft}$$

Some measurements of angles of $TUVW$ and $TZYX$ are given.

$$\begin{array}{l} m\angle VUZ = (12c - 6)^\circ, m\angle YZT = (10c + 5)^\circ, m\angle XTZ = 90^\circ, \\ m\angle TXY = (4d + 10)^\circ, m\angle TWV = (5d - 10)^\circ \end{array}$$

9. Determine the value of each variable:

$$\begin{array}{lcl} c = \underline{\frac{11}{2}} & 12c - 6 = 10c + 5 & 4d + 10 = 5d - 10 \\ d = \underline{20} & 2c = 11 & d = 20 \\ & c = \frac{11}{2} & \end{array}$$

10. What is the measure of $\angle XYZ$?

$$\begin{array}{lcl} 10\left(\frac{11}{2}\right) + 5 = 55 + 5 = 60^\circ & 60 + 90 + 90 = 240^\circ & \\ 4(20) + 10 = 80 + 10 = 90^\circ & 360^\circ - 240^\circ = 120^\circ & \end{array}$$

$$m\angle XYZ = 120^\circ$$

Geometry

Unit 6

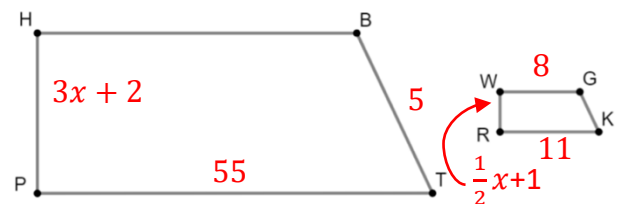
Lesson 13 Homework

Name Answer Key

Date _____

Period _____

Sheena, a party planner, is designing a banner for a birthday party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.



Sheena knows that $PT = 55$ inches (in), $HP = 3x + 2$ in, $WG = 8$ in, $WR = 0.5x + 1$ in, $BT = 5$ in, and $RK = 11$ in.

- What scale factor will map $HBTP$ to $WGKR$?

$$\frac{11}{55} = \boxed{\frac{1}{5}}$$

- What is the perimeter of $HBTP$, the main entrance banner?

$$\frac{3x+2}{\frac{1}{2}x+1} = \frac{5}{1}$$

$$3x + 2 = \frac{5}{2}x + 5$$

$$HP = 3(6) + 2$$

$$\frac{HB}{8} = \frac{5}{1}$$

$$3x + 2 = 5\left(\frac{1}{2}x + 1\right)$$

$$\frac{1}{2}x = 3$$

$$HP = 20$$

$$HB = 40$$

$$x = 6$$

$$P = 20 + 40 + 5 + 55 = \boxed{120 \text{ in}}$$

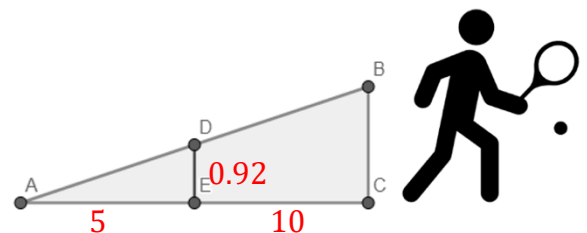
- What is the perimeter of $WGKR$, the photo booth banner?

$$\frac{120}{P} = \frac{5}{1}$$

$$5P = 120$$

$$\boxed{P = 24 \text{ in}}$$

Venus is doing a tennis drill where she wants to hit the ball so that it will just pass over the net, $\overline{DE}$, and land 5 meters (m) away from the base of the net.



- Given that $\triangle ADE \sim \triangle ABC$, $AE = 5$ m, $DE = 0.92$ m, and $EC = 10$ m, what is the height, represented by $\overline{BC}$, of the tennis racket?

$$\frac{15}{5} = \frac{BC}{0.92}$$

$$BC = 0.92\left(\frac{15}{5}\right)$$

$$\boxed{BC = 2.76 \text{ m}}$$

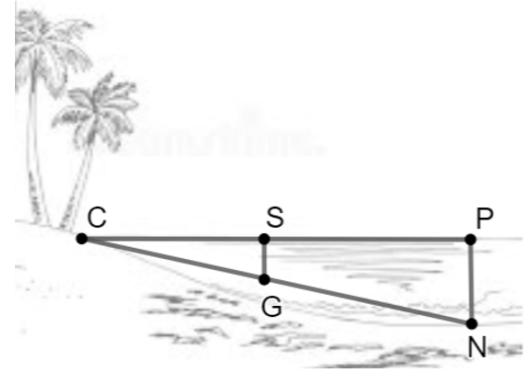
- What scale factor will map $\triangle ABC$ to $\triangle ADE$?

$$\frac{AE}{AC} = \frac{5}{15} = \boxed{\frac{1}{3}}$$

A beach front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SG}$ are the two main support posts for the dock.

$\triangle CSG \sim \triangle CPN$, $CS = 8$ feet (ft), $SG = (2x - 4)$ ft,

$PN = (5x + 8)$ ft, and $CP = 32$ ft.



6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?

$$\frac{CS}{CP} = \frac{8}{32} = \boxed{\frac{1}{4}}$$

7. What is the depth of the longer support post, represented by $\overline{PN}$?

$$\frac{5x+8}{2x-4} = 4$$

$$3x = 24$$

$$PN = 5(8) + 8$$

$$5x + 8 = 8x - 16$$

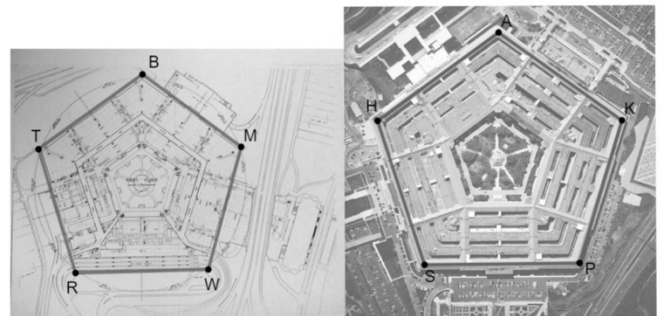
$$x = 8$$

$$\boxed{PN = 48 \text{ ft}}$$

8. What is the depth of the shorter support post, represented by $\overline{SG}$?

$$SG = 2(8) - 4 = \boxed{12 \text{ ft}}$$

Cesar wants to create a scale model of the Pentagon for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{5}$ of the scale model Pentagon, $AKPSH$. $TB = \left(\frac{2}{3}x + 4\right)$ centimeters (cm) and $HA = (6x + 4)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?

$$\frac{\frac{2}{3}x+4}{6x+4} = \frac{1}{5}$$

$$\frac{8}{3}x = 16$$

$$TB = \frac{2}{3}(6) + 4$$

$$P = 5 \cdot 8 \text{ cm}$$

$$\frac{10}{3}x + 20 = 6x + 4$$

$$x = 6$$

$$TB = 8 \text{ cm}$$

$$\boxed{P = 40 \text{ cm}}$$

10. What is the perimeter of the scale model Pentagon, $AKPSH$?

$$\frac{P}{40} = \frac{5}{1}$$

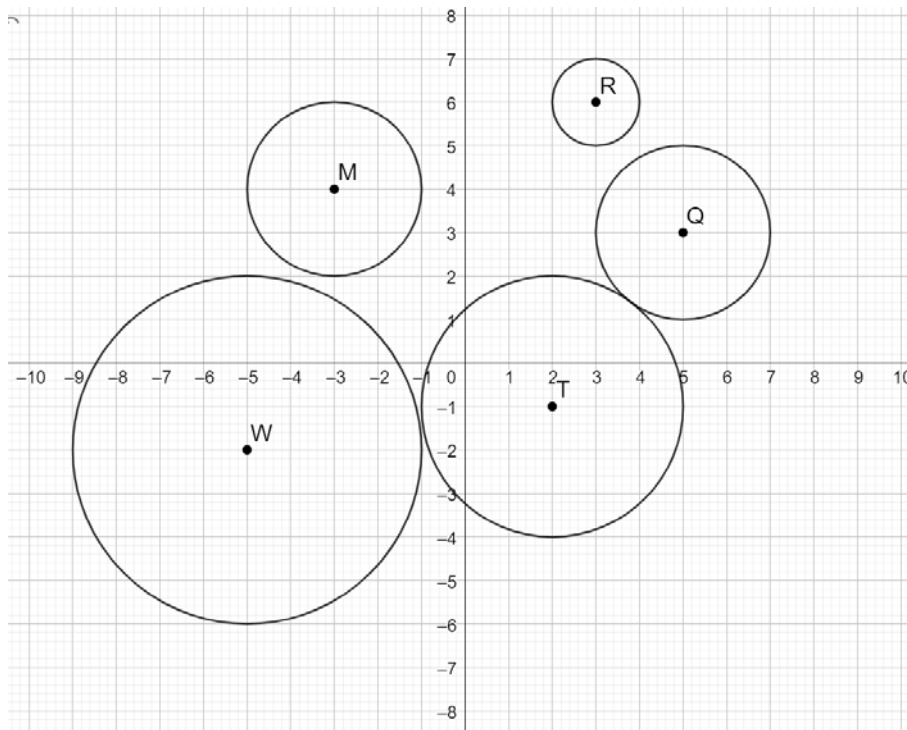
$$\boxed{P = 200 \text{ cm}}$$

Geometry
Unit 6
Lesson H1 Homework

Name Answer Key
Date _____
Period _____

1. _____ circles are similar.

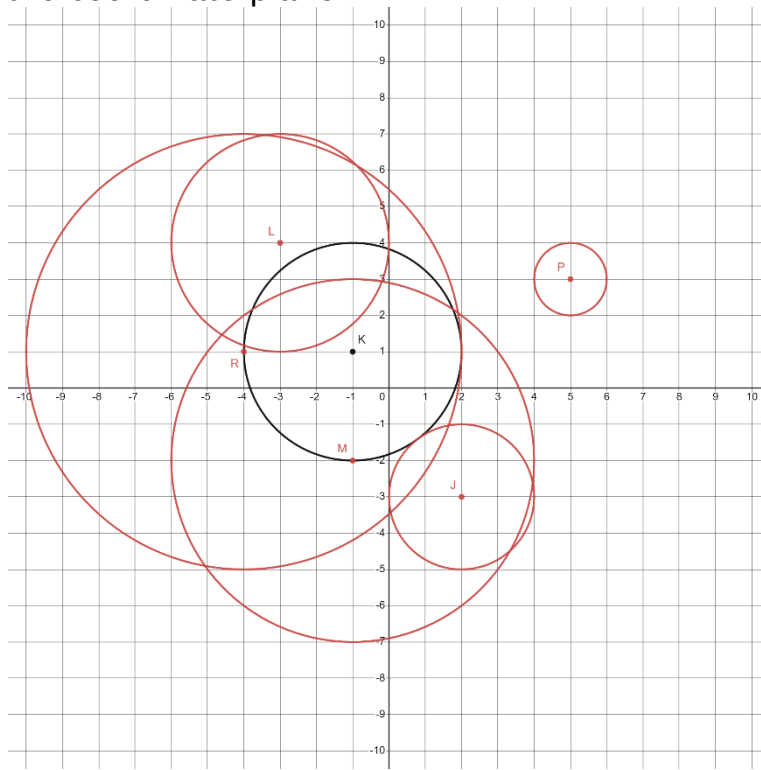
☒ All
☐ Some
☐ No



Given Circle M as the preimage:

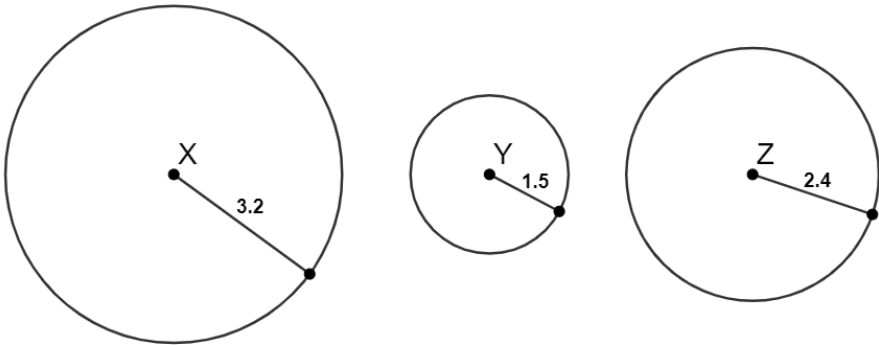
2. How can Circle M be mapped onto Circle R ?
Translate 6 units right and 2 units up and dilate with a scale factor of $\frac{1}{2}$ about the center.
3. How can Circle M be mapped onto Circle Q ?
Translate 8 units right and 1 unit down.
4. How can Circle M be mapped onto Circle T ?
Translate 5 units right and 5 units down and dilate with a scale factor of $\frac{3}{2}$ about the center.
5. How can Circle M be mapped onto Circle W ?
Translate 2 units left and 6 units down and dilate with a scale factor of 2 about the center.

Circle K is shown on the coordinate plane.



6. Draw Circle J , the result of translation 3 units right and 4 units down and a dilation of $\frac{2}{3}$ about the center of Circle K .
7. Draw Circle L , the result of translation 2 units left and 3 units up.
8. Draw Circle M , the result of a dilation of $\frac{5}{3}$ about the center of Circle K and a translation 3 units down.
9. Draw Circle P , the result of translation 6 units right and 2 units up and a dilation of $\frac{1}{3}$ about the center of Circle K .
10. Draw Circle R , the result of translation 3 units left and a dilation of 2 about the center of Circle K .

Circles X , Y , and Z are shown. The radius of Circle X is 3.2 units long. The radius of Circle Y is 1.5 units long. The radius of Circle Z is 2.4 units long.



1. Complete the following table. Round to the nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference
X	3.2	20.106
Y	1.5	9.425
Z	2.4	15.080

2. Complete the table.

Circle X and Y	
Ratio of Radius of Circle X to Radius of Circle Y	$\frac{3.2}{1.5} = 2.133$
Ratio of Circumference of Circle X to Circumference of Circle Y	$\frac{20.106}{9.425} = 2.133$

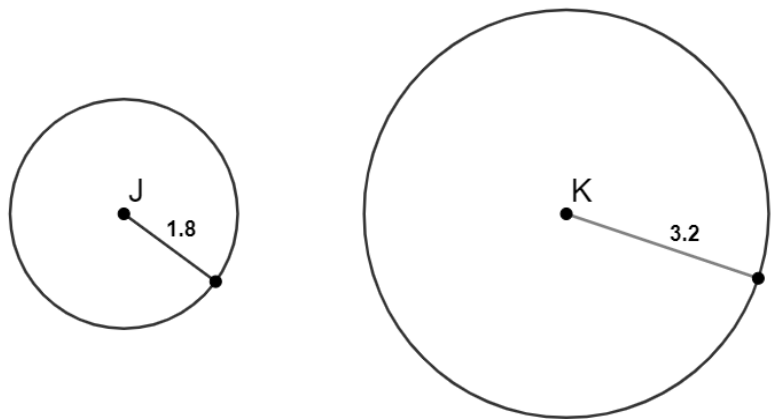
3. Complete the table.

Circle Y and Z	
Ratio of Radius of Circle Y and Radius of Circle Z	$\frac{1.5}{2.4} = 0.625 = \frac{5}{8}$
Ratio of Circumference of Circle Y and Circumference of Circle Z	$\frac{9.425}{15.080} = 0.625 = \frac{5}{8}$

4. Complete the table.

Circle X and Z	
Ratio of Radius of Circle X to Radius of Circle Z	$\frac{3.2}{2.4} = 1.333$
Ratio of Circumference of Circle X to Circumference of Circle Z	$\frac{20.106}{15.080} = 1.333$

Circles *J* and *K* are shown. The radius of Circle *J* is 1.8 units long. The radius of Circle *K* is 3.2 units long.



Show Circle *J* is similar to Circle *K* by completing the proof by contradiction:

- 5. Assume that Circle *J* is **not similar** to Circle *K*.
- 6. The ratio of the radius of Circle *J* to the radius of Circle *K* is **0.563**.
- 7. The ratio of the circumference of Circle *J* to the circumference of Circle *K* is **0.563**.
- 8. The ratio of the radii and the ratio of the circumferences are **equal**, so corresponding parts of Circle *J* and Circle *K* are proportional.
- 9. Since the corresponding parts of Circle *J* and Circle *K* are proportional, the circles are **similar**. This is a contradiction, so the assumption that Circle *J* is not similar to Circle *K* is not true.

10.

✖ All

Some

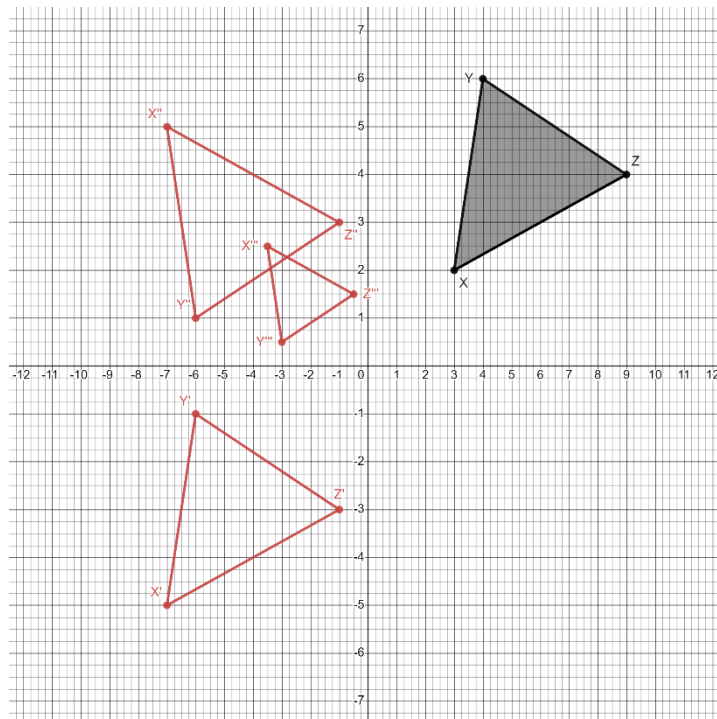
No

 circles are similar.

Geometry
Unit 6
Lesson H3 Homework

Name Answer Key
Date _____
Period _____

$\triangle XYZ$ is shown.



Draw the result of each transformation in the following sequence:

1. $\triangle X'Y'Z'$ as a result of $(x, y) \rightarrow (x - 10, y - 7)$
2. $\triangle X''Y''Z''$ as a result of $(x, y) \rightarrow (x, -y)$
3. $\triangle X'''Y'''Z'''$ as a result of $(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$

Indicate the correct relationship for the resulting image:

4. Image $\triangle X'Y'Z'$ is

☒ congruent
☐ similar

 to preimage $\triangle XYZ$.
5. Image $\triangle X''Y''Z''$ is

☒ congruent
☐ similar

 to preimage $\triangle XYZ$.
6. Image $\triangle X'''Y'''Z'''$ is

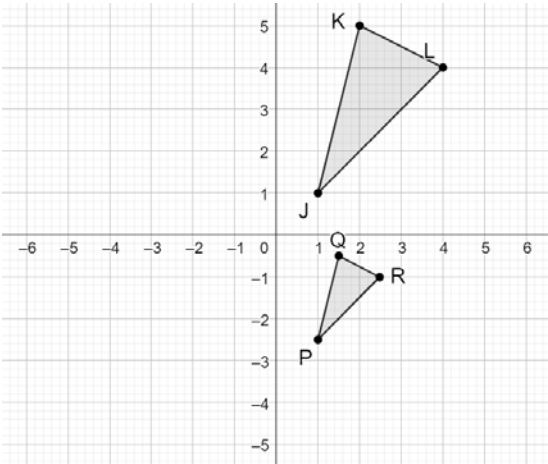
☐ congruent
☒ similar

 to preimage $\triangle XYZ$.

$\triangle JKL$ and $\triangle PQR$ are shown.

Assume: $\frac{JK}{PQ} = \frac{KL}{QR} = \frac{LJ}{RP}$

7. Write a series of transformations that will map $\triangle JKL$ onto $\triangle PQR$ so that $\triangle JKL \sim \triangle PQR$.
 $(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$ then $(x,y) \rightarrow (x+\frac{1}{2},y-3)$

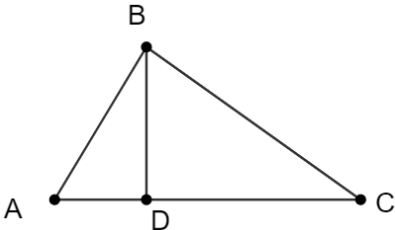


8. If it is given that $\triangle PQR$ is a result of $(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$ then $(x,y) \rightarrow (x+\frac{1}{2},y-3)$,
then the triangles are

☐ congruent
☒ similar

.

$\triangle ABC$ is shown with $\overline{BD}$.



Complete the table:

	If Given	True or False? $\triangle ABC \sim \triangle ADB$ by AA Similarity	Justification
9.	$\overline{AB} \perp \overline{BC}$ and $\overline{BD} \perp \overline{AC}$	☒ True ☐ False	$\angle ABC \cong \angle ADB, \angle BAC \cong \angle DAB$
10.	$\frac{AB}{AD} = \frac{BC}{BD}$	☐ True ☒ False	Two sides are proportional, but no angles nor the third side are given in order to confirm similarity.